

EMPLOYEE ATTRITION ANALYSIS

1. Introduction

Employee attrition refers to employees leaving an organization. High employee attrition can create challenges for an organization because it may result in recruitment and training requirements, productivity disruption, loss of organizational knowledge, and additional workload for remaining employees.

This project studies employee attrition using an HR employee dataset. The objective is to understand the business problem, examine the available HR features, formulate attrition as a prediction problem, perform basic exploratory data analysis, and plan a future machine learning workflow that can support employee-retention decisions.

2. Business Understanding

2.1 Business Case

The organization needs to understand patterns associated with employees leaving so that HR teams and managers can make better-informed retention decisions. Employee attrition can increase recruitment and onboarding requirements and may affect productivity and workforce continuity.

2.2 Business Problem Statement

The purpose of this project is to analyze employee HR data and identify patterns associated with employee attrition. The project will examine demographic, job-related, satisfaction, compensation, experience, overtime, and other HR-related characteristics to support the prediction of whether an employee is likely to leave the organization.

2.3 Project Objective

- Understand the employee attrition business problem.
- Analyze the available HR employee dataset.
- Identify factors and patterns associated with attrition.
- Formulate employee attrition as a binary classification problem.
- Define business and model evaluation KPIs.
- Consider ethical issues related to employee-level prediction.
- Create a structured workflow for future modeling and analysis.

2.4 HR Analytics

HR analytics uses employee-related data to understand workforce patterns and support HR decision-making. In this project, HR analytics is applied to employee attrition by

examining employee demographics, job characteristics, satisfaction, compensation, work experience, overtime, training, and career progression.

2.5 Employee Retention Strategies

- Improve employee satisfaction and engagement.
- Review overtime and workload patterns.
- Provide career-development and training opportunities.
- Support healthy work-life balance.
- Improve communication between employees and managers.
- Use department- and role-specific retention initiatives based on observed patterns.

2.6 Business Stakeholders

Stakeholder	Interest / Responsibility
HR Department	Employee retention, workforce analysis and HR strategy
HR Managers	Monitor attrition and identify retention opportunities
Department Managers	Understand attrition patterns within teams
Senior Management	Make strategic workforce decisions
Finance Department	Understand recruitment and replacement cost implications
Employees	Benefit from improved workplace and retention initiatives
Data Science Team	Analyze data and develop predictive models

2.7 Impact of Attrition

- Recruitment and hiring costs.
- Training and onboarding requirements.
- Loss of experienced employees and organizational knowledge.
- Potential reduction in productivity during replacement periods.
- Increased workload for remaining employees.
- Possible effects on team stability and morale.

3. Dataset Understanding

3.1 Dataset Summary

Dataset Property	Result
Number of records	1,470
Number of columns	35
Integer columns	26
Object/categorical columns	9

Target variable	Attrition
Missing values	0
Duplicate rows	0
Attrition = No	1,233
Attrition = Yes	237
Approximate attrition rate	16.12%

3.2 Target Variable

The target variable is Attrition. It contains two classes: Yes, indicating employee attrition, and No, indicating that the employee did not leave. The dataset contains 237 employees with Attrition = Yes and 1,233 with Attrition = No. Therefore, the problem is formulated as a binary classification task.

3.3 Feature Dictionary

Feature	Type	Description
Age	Numerical	Age of the employee
Attrition	Target / Categorical	Indicates whether the employee left
BusinessTravel	Categorical	Employee business-travel category
DailyRate	Numerical	Employee daily rate
Department	Categorical	Department in which the employee works
DistanceFromHome	Numerical	Distance from employee's home to workplace
Education	Ordinal	Employee education level
EducationField	Categorical	Employee's field of education
EmployeeCount	Constant / Numerical	Employee count field
EmployeeNumber	Identifier	Employee identification number
EnvironmentSatisfaction	Ordinal	Satisfaction with work environment
Gender	Categorical	Employee gender
HourlyRate	Numerical	Employee hourly rate
JobInvolvement	Ordinal	Level of involvement in the job
JobLevel	Ordinal	Employee job level
JobRole	Categorical	Employee's job role
JobSatisfaction	Ordinal	Employee job-satisfaction level
MaritalStatus	Categorical	Employee marital-status category
MonthlyIncome	Numerical	Employee monthly income
MonthlyRate	Numerical	Employee monthly rate

NumCompaniesWorked	Numerical	Number of companies previously worked for
Over18	Categorical	Whether employee is over 18
OverTime	Categorical	Whether employee works overtime
PercentSalaryHike	Numerical	Percentage increase in salary
PerformanceRating	Ordinal	Employee performance rating
RelationshipSatisfaction	Ordinal	Satisfaction with workplace relationships
StandardHours	Constant / Numerical	Standard working-hours field
StockOptionLevel	Ordinal	Employee stock-option level
TotalWorkingYears	Numerical	Total years of work experience
TrainingTimesLastYear	Numerical	Training sessions attended during previous year
WorkLifeBalance	Ordinal	Employee work-life-balance rating
YearsAtCompany	Numerical	Years employee has worked at the company
YearsInCurrentRole	Numerical	Years in current role
YearsSinceLastPromotion	Numerical	Years since last promotion
YearsWithCurrManager	Numerical	Years working with current manager

4. Basic Exploratory Data Analysis

4.1 Missing Values

The dataset contains no missing values. All 35 columns contain 1,470 non-null observations. Therefore, missing-value imputation is not required at this stage.

4.2 Duplicate Records

The dataset contains zero duplicate rows. No duplicate-row removal is required based on the current dataset check.

4.3 Attrition Distribution

There are 1,233 employees with Attrition = No and 237 employees with Attrition = Yes. The resulting attrition rate is approximately 16.12%. The target is therefore not evenly distributed, which should be considered during future model evaluation.

4.4 Department-wise Attrition Observation

Department	No (%)	Yes (%)
Human Resources	80.95%	19.05%
Research & Development	86.16%	13.84%
Sales	79.37%	20.63%

The observed attrition percentages differ across departments. Sales has the highest observed attrition percentage among the three departments shown, while Research & Development has the lowest. These values indicate association in the dataset and should not be interpreted as proof that department causes attrition.

4.5 Correlation Analysis

A correlation matrix was included in the notebook as part of the basic EDA. Correlation analysis can help identify relationships among numerical variables and may support feature selection and interpretation during later modeling.

5. Problem Formulation

5.1 Prediction Problem

The employee attrition problem is formulated as a supervised binary classification task. The model will use available employee features to predict whether Attrition is Yes or No.

5.2 Business Question

Can employee HR information be used to predict which employees are likely to leave the organization?

5.3 Business KPIs

- Overall Attrition Rate = $\text{employees who left} / \text{total employees} \times 100$.
- Department-wise Attrition Rate.
- Job-role Attrition Rate.
- Overtime-related Attrition Rate.
- Job Satisfaction versus Attrition.
- Employee Retention Rate.
- High-risk employee identification rate.

5.4 Model Evaluation KPIs

- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

Because the attrition classes are imbalanced, accuracy alone should not be treated as the only model-performance measure.

6. Ethical Considerations

Privacy

Employee information should be protected and accessed only by authorized users.

Fairness

The model should be evaluated for potentially unfair differences across demographic groups.

Bias

Historical HR data may contain historical organizational patterns or biases that a model could reproduce.

Transparency

Predictions should be treated as risk indicators rather than certain statements about an employee.

Human Oversight

HR decisions should not be made solely from an automated prediction.

Responsible Use

Predictions should support constructive retention actions such as engagement, development and support, rather than punishment or automatic termination.

7. Project Workflow

1. Business Understanding
2. Dataset Collection
3. Dataset Understanding
4. Data Quality Check
5. Exploratory Data Analysis
6. Feature Engineering
7. Data Preprocessing
8. Train/Test Split
9. Model Building
10. Model Evaluation
11. Attrition Prediction

12. Business Insights

13. Employee Retention Strategies

8. Project Roadmap

Phase	Activity	Main Output
Phase 1	Business Understanding	Define business problem, stakeholders and impact
Phase 2	Dataset Understanding	Review features, target and data quality
Phase 3	EDA	Analyze distributions, attrition and correlations
Phase 4	Preprocessing	Encode categorical variables and prepare data
Phase 5	Modeling	Train classification models
Phase 6	Evaluation	Evaluate using appropriate classification metrics
Phase 7	Insights	Identify important patterns and business findings
Phase 8	Reporting	Document results and prepare GitHub submission

9. GitHub Repository Plan

Recommended repository structure:

- README.md
- Employee_Attrition_Analysis (1).ipynb
- Feature_Dictionary.md
- Dataset_Report.md
- Business_Problem_Statement.md
- Ethics_Report.md
- requirements.txt
- data/ (only if redistribution is permitted)

Before publishing the dataset in a public GitHub repository, confirm that the dataset may be redistributed publicly. If redistribution is not permitted, keep the data file out of the public repository.

10. Conclusion

This project defines employee attrition as a business and analytics problem. The available dataset contains 1,470 employee records and 35 columns, with Attrition as the

binary target. Initial checks show no missing values and no duplicate records. The basic EDA indicates that approximately 16.12% of employees in the dataset are classified as attrited, and department-level attrition percentages differ. The next stages of the project are preprocessing, feature engineering, model development, evaluation, and translation of model findings into responsible employee-retention insights.